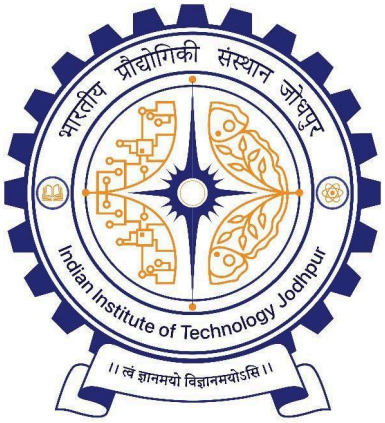
A new specialized PG degree program in the Department of Materials Engineering

M.TECH.in Battery Materials & Technology



Department of Materials Engineering

Indian Institute of Technology Jodhpur

2026-27

Concept Note

Background

1. Objectives of the Program

The global transition toward electrification, renewable energy integration, and sustainable mobility has created an unprecedented demand for advanced energy storage systems. Batteries are central to electric vehicles, grid-scale storage, portable electronics, aerospace, and emerging applications such as drones and wearables. India's ambitious targets for electric mobility, renewable energy penetration, and domestic manufacturing under initiatives like **National Electric Mobility Mission, Advanced Chemistry Cell (ACC) & Auto PLI Scheme**, and **Make in India** further intensify the need for a highly skilled workforce in battery technologies.

While conventional engineering programmes touch upon electrochemistry or power systems in isolation, there is a critical gap in interdisciplinary training that integrates materials science, electrochemistry, thermal management, electronics, manufacturing, safety, and lifecycle analysis of batteries. An M.Tech.programme in Battery Technologies is therefore proposed to develop industry-ready professionals, researchers, and innovators capable of addressing current and future challenges in energy storage.

The proposed M.Tech.programme in Battery Technologies addresses a critical national and global need for skilled manpower in advanced energy storage. By integrating fundamentals, hands-on training, and industry-relevant research, the programme will significantly contribute to technological innovation, industrial growth, and sustainable development.

2. Graduate Attributes

Graduates of the programme will be able to:

1. Understand fundamental electrochemical principles and battery chemistries.
2. Develop and characterize novel battery materials
3. Design, fabricate and characterize battery cells/packs.
4. Analyse battery performance, degradation, safety, and reliability.
5. Develop battery management systems and thermal management strategies.
6. Apply sustainability concepts including recycling, second-life usage, and lifecycle assessment.

3. Learning Outcomes

On successful completion of the programme, students will be able to:

1. Select appropriate battery chemistries for specific applications.
2. Develop novel battery materials for high-performance batteries.
3. Design battery packs integrated with power electronics and control systems.
4. Perform electrochemical, structural, thermal, and safety characterization of batteries.
5. Model battery behaviour and degradation using analytical and computational tools.
6. Address safety, environmental, and regulatory aspects of battery systems.

Course Structure:

Semester-wise Distribution of Credits

Cat.	Course Number, Course Title		L-T-P	Credits		Cat.	Course Number, Course Title		L-T-P	Credits
I Semester						**II Semester**				
C	MTL7XX0	Battery Materials Science	3-0-0	3		**C**	MTL7XX0	Computational Electrochemistry and AI-driven Battery Design & Diagnostics	3-0-0	3
C	MTL7XX0	Battery Cell Fabrication and Manufacturing Technology	3-0-0	3		**C**	MTL7XX0	Battery Regulatory & Safety Practices	3-0-0	3
C	MTL7XX0	Battery Material Synthesis and Device Testing Lab	0-0-6	3		**E**	MTL7XX0	Elective-2	3-0-0	3
E	MTL7XX0	Elective-1	3-0-0	3		**E**	MTL7XX0	Elective-3	3-0-0	3
NG1	LAN7XX0	Technical Communication	1-0-0	S/U		**NG2**	OAL7XX0	Innovation and IP Management	1-0-0	S/U
			Total	12					Total	12
Summer M.Tech.Thesis										
T	MTT7XX0	Summer Thesis								8
III Semester						**IV Semester**				
T	MTT7XX0	Thesis		16		T	MTT7XX0	Thesis		16
			Total	16					Total	16

Program's Structure

S.No.	Course Type	Credits
1	Compulsory	15
2	Program Electives	9
3	M.Tech.Thesis I	16
4	M.Tech.Thesis II	16
5	M.Tech.Thesis summer	8
Graded		**64**
6	Non-Graded Courses (2 Credits)	0
Total		**64**

Electives Courses:

S.No.	Course Code	Course Title	L-T-P	Credits
1	MTL7280	Materials for Electrochemical Energy Conversion and Storage	3-0-0	3
2	MTL7XX0	Batteries with Li and Beyond-Li Chemistry	3-0-0	3
3	MTL7XX0	Solid-State Batteries	3-0-0	3
4	MTL7XX0	Battery-Based Hybrid Energy Storage Systems	3-0-0	3
5	MTL7XX0	Advanced Battery Chemistry for Large-Scale Energy Storage	3-0-0	3
6	MTL7XX0	Batteries for Portable Electronics	3-0-0	3
7	MTL7XX0	Stationary Batteries & Battery-Grid Integration	3-0-0	3
8	MTL7XX0	Traction Batteries: Battery Materials & Pack Design	3-0-0	3
9	MTL7XX0	Battery Management Systems	3-0-0	3
10	MTL7XX0	Battery Lifecycle Assessment	3-0-0	3

Compulsory Courses

Title	Battery Materials Science	Number	MTL7XX0
Department	Materials Engineering	L-T-P [C]	3-0-0 [3]
Offered for	M.Tech.(BMT) / M.Sc. / B.Tech. (Final Year)/Ph.D.	Type	C
Prerequisite			

Objective:

The Instructor will

1. Introduce the fundamental materials and electrochemical principles underlying modern battery cells, covering electrode and electrolyte materials, interfacial phenomena.
2. Explain degradation mechanisms that govern battery performance, energy density, and lifetime.

Learning Outcomes:

3. Identify and compare cathode, anode, and electrolyte materials for various battery chemistries.
4. Explain electrochemical working principles of Li-ion and beyond-Li battery systems.

Contents:

Fundamentals of Electrochemical Energy Storage Thermodynamics of electrochemical cells; electrode potential and Nernst equation; Faradaic and non-Faradaic processes; cell voltage, capacity, energy density and power density; Butler-Volmer equation for electrochemical kinetics; overpotentials and their origins.

Anode Materials: Graphite structure and lithiation mechanism; silicon and tin-based alloy anodes; lithium titanate (LTO) anodes; lithium metal anodes and dendrite challenges; sodium and multivalent metal anodes; conversion-type anode materials.

Cathode Materials: Layered oxide cathodes: LCO, NMC, NCA; spinel cathodes: LMO; olivine cathodes: LFP; high-nickel cathode chemistry; sodium-ion cathodes: layered oxides, Prussian blue analogues; structural stability and capacity degradation; high-voltage cathode materials.

Electrolyte Materials: Liquid electrolytes: solvents and lithium salts; ionic conductivity and electrochemical stability window; solid polymer electrolytes; inorganic solid electrolytes: oxide and sulfide types; Solid Electrolyte Interphase (SEI) formation and stability; electrolyte additives and their functions.

Separators, Current Collectors, and Cell Architecture: Separator materials and properties; ceramic-coated separators; current collector materials and surface treatments; cell formats: cylindrical, pouch, prismatic; pack architecture and design considerations.

Degradation and Failure Mechanisms: Capacity fade mechanisms in Li-ion batteries; lithium plating and stripping phenomena; mechanical degradation: cracking and delamination; thermal aging and electrolyte decomposition; calendar and cycle aging models; failure analysis techniques.

Text Books

1. Wu, Y. (2015). *Lithium-Ion Batteries: Fundamentals and Applications*. CRC Press.
2. Newman, J., & Thomas-Alyea, K. E. (2004). *Electrochemical Systems* (3rd ed.). Wiley.

Reference:

1. Basu Majumder, S. Electrochemical Energy Storage. NPTEL. https://nptel.ac.in/courses/113105102
2. Bazant, M. (2014). Electrochemical Energy Systems (10.626). MIT OpenCourseWare. https://ocw.mit.edu/courses/10-626-electrochemical-energy-systems-spring-2014

Title	Battery Cell Fabrication and Manufacturing Technology	Number	MTL7XX0
Department	Materials Engineering	L-T-P [C]	3-0-0 [3]
Offered for	M.Tech.(BMT) / M.Sc. / B.Tech. (Final Year)/Ph.D.	Type	C
Prerequisite	Battery Materials Science		

Objective:

The Instructor will

1. Provide comprehensive understanding of the manufacturing processes involved in battery cell and pack production.
2. Cover electrode preparation, cell assembly, formation cycling, quality control, and scale-up strategies for industrial battery manufacturing.

Learning Outcomes:

The students are expected to have the ability to

1. Describe each step in battery cell and pack manufacturing and the process-structure-property relationships in electrode fabrication.
2. Assess quality control methods and evaluate manufacturing challenges associated with scale-up and cost reduction.

Contents:

Battery Manufacturing Overview: Battery manufacturing landscape in India and globally; cost breakdown of Li-ion battery production; manufacturing process flow for cylindrical, pouch, and prismatic cells; raw material sourcing and supply chain; key equipment and facility requirements.

Electrode Fabrication: Slurry preparation: binders, solvents, conductive additives; coating technologies: slot-die, gravure, and blade coating; drying and calendaring processes; electrode porosity and density optimization; dry electrode processing (solvent-free methods); current collector treatment; quality metrics for electrodes: uniformity, adhesion, density.

Cell Assembly: Electrode slitting and notching; winding and stacking processes; separator integration and alignment; electrolyte filling and impregnation; cell sealing and can crimping; dry room and controlled atmosphere requirements; glove box and moisture control practices.

Formation and Aging: Formation cycling protocols; SEI formation and its role in performance; degassing processes; aging and grading of cells; self-discharge measurement; capacity and voltage matching for pack assembly.

Battery Pack Assembly and Testing: Module and pack architecture design; busbars, connectors, and structural components; thermal management integration; BMS hardware integration; pack-level electrical testing; safety testing standards: AIS-156, IEC 62133; end-of-line quality assurance.

Scale-Up and Manufacturing Challenges: Pilot-scale to gigafactory scale-up strategies; yield improvement and defect root cause analysis; digital manufacturing and Industry 4.0 in battery production; environmental and safety aspects; recycled material integration; future manufacturing trends: solid-state cell production.

Text books:

1. Kiehne, H. A. (Ed.). *Battery Technology Handbook*. Marcel Dekker.
2. Linden, D. (Ed.). *Handbook of Batteries*. McGraw-Hill.

Reference:

1. Jhunjhunwala, A. *Electric Vehicle Battery Technology*. IIT Madras, NPTEL. https://archive.nptel.ac.in/courses/108/106/108106182/

Title	Battery Material Synthesis and Device Testing Lab	Number	MTL7XX0
Department	Materials Engineering	L-T-P [C]	0-0-6 [3]
Offered for	M.Tech. (BMT) / M.Sc. / B.Tech. (Final Year)/Ph.D.	Type	C
Prerequisite	Battery Materials Science (co-requisite)		

Objective:

The Instructor will

1. Provide hands-on training in the synthesis of electrode and electrolyte materials and the use of standard characterization techniques employed in battery research and development.
2. Demonstrate analyse experimental data using standard battery research protocols.

Learning Outcomes:

The students are expected to have the ability to

1. Synthesize cathode and anode materials using solid-state and wet-chemical routes and characterize their phase, morphology, and electrochemical properties.
2. Operate and interpret results from key characterization instruments: XRD, SEM, BET, TGA, and electrochemical workstations.

Contents:

Introductory Battery Components Identification: Cathode materials, Anode materials, Separators, Electrolytes, Current collectors.
Synthesis of Electrode & Electrolyte Materials: Hard Carbon Preparation; Biomass-derived carbon synthesis; Metal-Based Anodes: Si-Graphite & Sb/Bi-hard carbon composite preparation by milling and mixing, Electrodeposited Metal-Based Anodes: Electrodeposited Sb/Bi as anode, Recovery of Anode Materials from Waste: Recovery of Zn from dead batteries and industrial waste.
Solid-State Synthesis: Preparation of layered oxide cathodes (e.g., $LiMO_2$), Solid Electrolyte Synthesis: Fabrication of ceramic & polymer film electrolyte, Recycling & Reuse: Recovery of lead and reuse in lead-acid batteries.
Characterization of Electrode & Electrolyte Materials: Structural Characterization; X-Ray Diffraction (XRD) of graphite, hard carbon, Si & Sb; Phase dentification; Crystallite size analysis, Raman Spectroscopy & FTIR Spectroscopy: Carbon structure analysis; Functional group analysis. Morphological Characterization: Scanning Electron Microscopy (SEM), EDS & Particle Size, Analysis of carbon and metallic materials, BET Surface Area Measurement: Porosity and surface area measurement of carbon materials.
Electronic Conductivity: Measurement of electronic conductivity of carbon and metallic electrode materials by 4- probe method, Ionic Conductivity: Measurement of ionic conductivity of electrode materials and solid electrolytes by impedance spectroscopy.
Electrode Fabrication: Mass Loading & Slurry Preparation, Active material loading calculation, Mixing active material, binder, and conductive additives; Selection of binder systems (PVDF, CMC, SBR).
Electrode Coating: Doctor blade coating on aluminium and copper foils; Controlling electrode thickness, Electrode Calendaring, Drying & Punching: Density optimization; Porosity control; Vacuum drying procedures; Solvent removal; Electrode cutting.
Cell Fabrication: Fabrication of Primary Batteries; Fabrication of voltaic cell, Daniel cell, gravity cell, metal-air battery.
Introduction to Coin Cell Components & Assembly: Cell hardware components; Spring, spacer, gasket roles; Assembly of CR2032 coin cells in glovebox.
Pouch Cell Assembly: Layer stacking methods; Sealing techniques; Fabrication of pouch cell lead-acid and Li-ion batteries.

Fabrication of Micro batteries: Fabrication of intermetallic & displacement micro batteries at room and high temperatures.
Battery Testing & Electrochemical Characterization:
Battery Charging Protocols, Charge–Discharge Operation & Cyclic Voltammetry (CV): Constant current (CC) & Constant voltage (CV) charging & discharging; Coulombic Efficiency Measurement; Redox reaction analysis.
Cyclability Testing: Long-term cycling performance of different batteries.
Rate Capability Testing: Performance at different C-rates of different batteries.
Electrochemical Impedance Spectroscopy (EIS): Internal resistance analysis; Charge transfer resistance.
Galvanostatic/Potentiostatic Intermittent Titration Technique (GITT & PITT): Diffusion coefficient estimation; Ion transport analysis.
Specific Capacity, Energy & Power Calculation: Specific capacity calculation; Gravimetric and volumetric energy density calculation.

Text Books

1. Gueye, A. B., Maria, H. J., Kalarikkal, N., Fall, M., Stephan, A. M., & Thomas, S. (Eds.). (2024). *Nanostructured Materials Engineering and Characterization for Battery Applications*. Elsevier.
2. Li, J., & Jin, C. (2023). *Processing and Manufacturing of Electrodes for Lithium-Ion Batteries*. The Institution of Engineering and Technology.
3. Bard, A. J., & Faulkner, L. R. (2001). *Electrochemical Methods: Fundamentals and Applications* (2nd ed.). Wiley.

Online Courses / Lectures

1. Grzywa, M., & Weigold, L. *Webinar: Preparation and Characterization of Battery Materials*. Rigaku & Retsch. https://www.rigaku.com/webinar/battery-prep
2. Kayyar, A., Huang, J., Samiee, M., & Luo, J. (2012). Construction and testing of coin cells of lithium-ion batteries. *Journal of Visualized Experiments*, 66, e4104. https://www.jove.com/t/4104
3. Michigan Engineering Online. *Li-ion Battery Manufacturing and Safety* (Professional Certificate Course). University of Michigan. https://ope.engin.umich.edu/pe/lithium-battery-mfg-safety

Title	Computational Electrochemistry and AI-driven Battery Design & Diagnostics	Number	MTL7XX0
Department	Materials Engineering	L-T-P [C]	3-0-0 [3]
Offered for	M.Tech.(BMT) / M.Sc. / B.Tech. (Final Year)/Ph.D.	Type	C
Prerequisite	Battery Materials Science		

Objective:

The instructor will provide an insight into:

1. Quantitative and computational description of electrochemical processes relevant to batteries (kinetics, transport, impedance, multiscale modelling).
2. Physics-based modelling and numerical simulation of battery cells and packs (equivalent-circuit and porous-electrode models), AI-driven (machine-learning and deep-learning) methods for battery state estimation, lifetime prediction, and diagnostics at the cell and system level.

Learning Outcomes:

The students are expected to have the ability to

3. Formulate and solve basic electrochemical models (diffusion, charge transfer, impedance) numerically for battery-relevant systems.
4. Develop and use equivalent-circuit and physics-based models (SPM, P2D) for cell performance and degradation analysis, Design data pipelines and apply ML/AI techniques for SOC/SOH estimation, prognostics, and AI-assisted battery design.

Contents:

Computational Electrochemistry Foundations: Chemical vs electrochemical kinetics; Butler-Volmer equation, Tafel behaviour; transport in electrolytes and electrodes (diffusion, migration), 1D model set-up; numerical solution of diffusion/kinetics boundary-value problems; introduction to voltammetry and impedance modelling (Randles circuit, Warburg).

Multiscale and DFT Perspectives for Batteries: Motivation for electronic-structure methods in batteries; conceptual introduction; typical DFT workflows and outputs; linking DFT-derived parameters to continuum battery models and databases.

Battery Modelling and Simulation: Brief recap of cell operation and definitions (SOC, SOH, C-rate) from a modelling perspective; equivalent-circuit models; parameter identification from pulse/HPPC tests; porous-electrode theory, single-particle model and pseudo-2D models; numerical solution strategies (discretization, time stepping, stiffness); simple degradation modeling (SEI growth, resistance increase) and validation against data.

AI-Driven Battery Design & Diagnostics: Overview of AI/ML for batteries (supervised learning, deep learning, physics-informed models); data sources and pipelines for cell and pack data; feature engineering from cycling and field datasets; SOC/SOH estimation with classical filters and ML regressors; deep time-series models (RNN/CNN/LSTM) and physics-informed neural networks as surrogates for battery models; AI-assisted design and lifetime prediction workflows.

Text books

1 Newman, J., & Thomas-Alyea, K. E. (2004). *Electrochemical Systems* (3rd ed.). Wiley.

2 Plett, G. L. (2015). *Battery Management Systems* (Vols. I–II). Artech House.

Self-learning

1. Thelen A. et al., Probabilistic machine learning for battery health diagnostics and prognostics. npj Materials Sustainability, volume 2, Article number: 14 (2024).
2. Bazant, M. (2014). Electrochemical Energy Systems (10.626). MIT Open Course Ware. https://ocw.mit.edu/courses/10-626-electrochemical-energy-systems-spring-2014

Title	Battery Regulatory & Safety Practices	Number	MTL7XX0
Department	Materials Engineering	L-T-P [C]	3-0-0 [3]
Offered for	M.TECH.(BMT) / M.Sc. / B.Tech. (Final Year)/Ph.D	Type	C
Prerequisite			

Objective:

1. To familiarise students with national and international battery safety standards including IEC, UN, and IS norms.
2. To develop the ability to identify, assess, and mitigate safety hazards across battery design, manufacturing, and end-of-life stages.

Learning Outcomes:

3. Evaluate battery safety risks and apply relevant standards and regulations to ensure safe battery operation and compliance.
4. Interpret and apply battery safety regulations to evaluate compliance in design, transportation, and storage applications.

Contents:

Introduction to Battery Safety and Regulations: Importance of safety in battery technologies, Historical battery failures and safety incidents, Safety challenges in modern battery systems, Overview of global battery regulations, Risk assessment in battery systems, Safety considerations across the battery lifecycle.

Battery Hazards and Failure Mechanisms: Electrochemical hazards in batteries, Thermal runaway mechanisms in lithium batteries, Overcharging and over-discharging risks, Internal short circuits and dendrite formation, Gas generation and pressure buildup, Mechanical damage and safety risks, Environmental hazards and toxic materials.

Safety Design and Protection Systems: Safety design principles for battery cells, Safety mechanisms in battery packs, Current interrupt devices and safety vents, Battery Management System (BMS) for safety, Thermal protection systems, Electrical isolation and protection circuits, Fail-safe design strategies.

Battery Safety Testing and Certification: Safety testing protocols for batteries, Abuse testing (thermal, mechanical, electrical), Transport safety testing, Performance and reliability testing, Fire safety and explosion prevention, Certification processes for batteries.

Battery Regulations and Standards: International battery safety standards (IEC 62133, UL 1642, UN 38.3), Regulations for lithium-ion batteries, Transportation regulations (ADR, IATA DGR, IMDG), Safety standards for electric vehicle batteries (AIS-156), Standards for stationary energy storage systems, Compliance and certification processes.

Battery Lifecycle Safety and Sustainability: Safe manufacturing practices, Safe storage and handling of batteries, Battery transportation and logistics safety, End-of-life management and recycling regulations, Environmental regulations and sustainability, Second-life battery safety considerations, Future safety technologies and regulatory trends.

Text books

1. Wang, Z., Ouyang, D., & Cai, Q. (2026). *Lithium-Ion Battery Safety*. Elsevier.
2. Pecht, M. G. (Ed.). (2025). *The Safety Challenges and Strategies of Using Lithium-Ion Batteries*. Wiley-IEEE Press.

Self-Learning

1. UNSW Digital Grid Futures Institute & UNSW TRaCE. (2024). *An Essential Guide to Lithium-Ion Battery Systems.* University of New South Wales. https://www.openlearning.com/unswshortcourses/courses/understanding-the-risks-of-lithium-ion-battery-systems

Elective Courses

Title	Batteries with Li and Beyond-Li Chemistry	Number	MTL7XX0
Department	Materials Engineering	L-T-P [C]	3-0-0 [3]
Offered for	M.Tech.(BMT) / M.Sc. / B.Tech. (Final Year)/Ph.D.	Type	E
Prerequisite			

Objective:

The Instructor will

1. Demonstrate electrochemical principles and material systems used in lithium-based and emerging beyond-lithium battery chemistry.
2. Introduce post-Li-ion chemistries (Na-ion, Mg, Zn-ion, Al-ion) and identify their key advantages, limitations, and suitability

Learning Outcomes:

1. Evaluate the synthesis, properties, and electrochemical performance of advanced Li-based batteries (Li metal, Li-S, Li-air) in terms of energy density, power density, and cycle life.
2. Compare post-Li-ion chemistries (Na-ion, Mg, Zn-ion, Al-ion) and identify their key advantages, limitations, and suitability for real-world energy storage applications.

Contents:

Fundamentals of Lithium Battery Chemistry: Evolution of lithium battery technology, Thermodynamics and electrochemical principles of lithium cells, Lithium metal vs lithium-ion battery chemistry, Structure and working principle of lithium-ion batteries, Intercalation chemistry in lithium battery electrodes, Electrolytes used in lithium batteries, Solid Electrolyte Interphase (SEI) formation and stability.

Lithium Battery Materials: Anode Materials: Graphite structure and lithiation mechanism, Silicon-based anodes and alloying chemistry, Lithium titanate (LTO) anodes, Lithium dendrite formation, Conversion and alloy-type lithium anodes; Cathode Materials: Layered oxide cathodes (LCO, NMC, NCA), Spinel cathodes (LMO), Olivine cathodes (LFP), High-nickel cathode chemistry, Structural degradation and cathode stability.

Advanced Lithium Battery Chemistries: Lithium-sulfur battery chemistry, Lithium-air (Li-O2) battery systems, Solid-state lithium batteries, Lithium polymer batteries, Electrolyte innovations for advanced lithium batteries, Safety and thermal stability of lithium systems.

Sodium and Other Alkali-Ion Batteries: Anode Materials: Hard carbon anodes for sodium-ion batteries, Alloy and conversion anodes in sodium systems; Cathode Materials: Layered oxide cathodes for sodium-ion batteries, Polyanionic cathodes (NASICON, Prussian blue), Potassium-ion battery chemistry, Magnesium-ion battery chemistry, Calcium-ion battery chemistry.

Multivalent and Metal Batteries: Anode Materials: Magnesium metal anodes, Aluminum metal anodes, Zinc metal anodes; Cathode Materials: Cathodes for multivalent batteries, Intercalation vs conversion cathode chemistry, Electrolyte challenges in multivalent batteries, Charge transport and reaction mechanisms.

Emerging Post-Lithium Battery Chemistries: Metal-air battery chemistries (Zn-air, Al-air, Fe-air), Metal-sulfur batteries beyond lithium, Organic electrode materials, Redox-flow battery chemistry, Liquid metal batteries.

Text books

1. Wu, Y. (2015). *Lithium-Ion Batteries: Fundamentals and Applications*. CRC Press.

2. Rajagopalan, R., Wang, H., & Tang, Y. (Eds.). (2024). *Advanced Metal Ion Storage Technologies: Beyond Lithium-Ion Batteries*. CRC Press (Taylor & Francis).

Reference books

1. Quartarone, E. (2021). *Recent Advances in Post-Lithium-Ion Batteries*. MDPI.
2. Pistoia, G. (2014). *Lithium-Ion Batteries: Advances and Applications*. Elsevier.

Title	Solid State Batteries	Number	MTL7XX0
Department	Materials Engineering	L-T-P [C]	3-0-0 [3]
Offered for	M.Tech.(BMT) / M.Sc. / B.Tech. (Final Year)/Ph.D.	Type	E
Prerequisite	Basic Electrochemistry / Materials Science		

Objective:

The Instructor will

1. Provide students with an in-depth understanding of solid-state battery technology.
2. Introduce to solid electrolyte materials, electrode–electrolyte interfaces, cell fabrication techniques, and the engineering challenges and opportunities for next-generation solid-state battery commercialization.

Learning Outcomes:

The students are expected to have the ability to

1. Compare oxide, sulfide, and polymer solid electrolyte systems on the basis of ionic conductivity, electrochemical stability, and processability, and analyse interfacial phenomena governing solid-state cell performance.
2. They will further evaluate fabrication strategies, mechanical and thermal challenges, and the commercial readiness of solid-state batteries for portable, traction, and stationary applications.

Contents

Introduction to Solid-State Batteries: Evolution of battery technologies and limitations of conventional lithium-ion batteries, Concept and principles of solid-state batteries, Comparison between liquid-electrolyte and solid-state batteries, Advantages and challenges of solid-state battery systems, Applications in electric vehicles, portable electronics, and grid storage, Current global research and development trends.

Solid Electrolyte Materials: Fundamentals of ion transport in solid electrolytes, Classification of solid electrolytes: ceramic, polymer, and composite, Oxide-based solid electrolytes (garnet, perovskite structures), Sulfide-based solid electrolytes, Polymer electrolytes and gel polymer systems, Ionic conductivity and transport mechanisms, Stability and compatibility with electrodes.

Electrode Materials for Solid-State Batteries: Cathode materials for solid-state batteries, High-voltage cathodes and interface compatibility, Lithium metal anodes, Alloy and composite anode materials, Interfacial reactions between electrodes and solid electrolytes, Interface engineering and coatings, Dendrite formation and suppression strategies.

Solid-State Battery Cell Design: Architecture of solid-state battery cells, Thin-film solid-state batteries, Bulk-type solid-state batteries, 3D structured solid-state batteries, Fabrication

techniques for solid-state cells, Interface engineering and stack design, Scaling challenges in solid-state battery manufacturing.

Performance, Safety, and Testing: Electrochemical characterization of solid-state batteries, Charge–discharge behaviour and rate performance, Thermal stability and safety advantages, Mechanical stability and stress management, Failure mechanisms and degradation, Reliability and cycle life testing.

Manufacturing, Applications, and Future Trends: Manufacturing technologies for solid-state batteries, Integration into electric vehicles and portable devices, Solid-state batteries for grid-scale storage, Cost and scalability challenges, Commercialization pathways and industry developments, Future directions in next-generation solid-state energy storage

Textbooks

1. Julien, C., & Nazri, G. A. (1994). *Solid State Batteries: Materials Design and Optimization*. Springer, Boston, MA.
2. Passerini, S., et al. (2024). *Solid State Batteries: Design, Challenges and Market Demands*. Springer Nature, Cham, Switzerland.

Reference Books

1. Dudney, N. J., West, W. C., & Nanda, J. (Eds.). (2015). *Handbook of Solid State Batteries* (2nd ed.). World Scientific Publishing, Singapore.
2. Daniel, C., & Besenhard, J. O. (Eds.). (2011). *Handbook of Battery Materials* (2nd ed.). Wiley-VCH, Weinheim, Germany.

Title	Battery-Based Hybrid Energy Storage Systems	Number	MTL7XX0
Department	Materials Engineering	L-T-P [C]	3-0-0 [3]
Offered for	M.Tech.(BMT) / M.Sc. / B.Tech. (Final Year)	Type	E
Prerequisite	Basic Electrochemistry / Materials Science		

Objectives:

1. Understand the fundamental principles of battery-based hybrid energy storage systems (HESS), including energy management, power sharing, and operational characteristics.
2. Study the different configurations of hybrid energy storage systems that integrate batteries with other storage technologies such as supercapacitors, fuel cells, and flywheels.

Learning Outcomes:

1. Analyze different hybrid energy storage system (HESS) architectures and their operational characteristics.
2. Evaluate the performance of hybrid energy storage systems in terms of efficiency, power management, and system reliability.

Contents:

Fundamentals of Hybrid Energy Storage Systems: Introduction to hybrid energy storage systems (HESS), Limitations of single battery systems, Need for hybridization in energy storage, Classification of hybrid energy storage systems, Electrochemical vs non-electrochemical storage devices, Power-energy characteristics of storage technologies, Applications of hybrid storage in modern energy systems.

Battery Technologies in Hybrid Systems: Role of batteries in hybrid storage systems, Lithium-ion battery chemistry and performance, Lead-acid batteries in hybrid configurations, Sodium-ion batteries for hybrid storage, Flow batteries in hybrid energy systems, Performance limitations of standalone batteries, Battery degradation in hybrid operation.

Battery-Supercapacitor Hybrid Systems: Fundamentals of supercapacitors, Electrochemical double layer capacitors (EDLC), Pseudocapacitors and hybrid capacitors, Battery-supercapacitor hybrid architecture, Power buffering and peak load management, Energy management strategies for battery-supercapacitor systems, Applications in electric vehicles and power electronics.

Battery Hybridization with Other Storage Technologies: Battery-fuel cell hybrid systems, Battery-flywheel hybrid storage, Battery-compressed air energy storage hybrids, Battery-hydrogen hybrid systems, Battery-thermal energy storage integration, Multi-storage hybrid architectures.

System Integration and Control: Power electronics for hybrid storage systems, DC-DC converters in hybrid energy systems, Energy management algorithms, Hybrid system modeling and simulation, Optimization of hybrid storage performance, Hybrid battery management systems (BMS).

Applications and Future Technologies: Hybrid storage for electric vehicles, Hybrid systems for renewable energy integration, Hybrid storage in smart grids, Hybrid energy storage for microgrids, safety and reliability of hybrid systems, Emerging hybrid energy storage technologies.

Textbooks

1. Design and Management of Energy-Efficient Hybrid Electrical Energy Storage Systems, Younghyun Kim and Naehyuck Chang.nger, Boston, MA.
2. Hybrid Renewable Energy Systems and Microgrids, Miloud Rezkallah, Elsevier

Reference Books

3. Hybrid Supercapacitor-Battery Energy Storage, Mainul Akhtar and S. B. Majumder, Springer
4. Power Management Control Strategy of Battery-Supercapacitor Hybrid Energy Storage System Used in Electric Vehicles, Qiao Zhang et al., Wiley-IEEE Press

Title	Advanced Battery Chemistry for Large-Scale Energy Storage	Number	MTL7XX0
Department	Materials Engineering	L-T-P [C]	3-0-0 [3]
Offered for	M.Tech.(BMT) / M.Sc. / B.Tech. (Final Year)	Type	EE
Prerequisite			

Objective:

1. Understand advanced electrochemical mechanisms involved in battery operation for large-scale energy storage applications.
2. Study the material chemistries used in batteries for large-scale energy storage systems.

Learning Outcomes:

1. Analyze the characteristics and performance of advanced battery chemistries used in energy storage systems.
2. Compare different battery chemistries to evaluate their suitability for grid-scale and stationary energy storage applications.

Contents:

Energy Storage Requirements for Grid Systems: Role of large-scale energy storage in modern power grids, Renewable energy intermittency and storage needs, Power vs energy requirements for grid applications, Cost metrics: $/kWh and $/cycle for grid batteries, Electrochemical fundamentals of large-scale storage, Thermodynamics and reaction kinetics in grid batteries, Safety and reliability in large battery installations.

Lithium-Based Chemistries for Stationary Storage: Anode Chemistry: Graphite intercalation chemistry for stationary Li-ion systems, Lithium titanate (LTO) anodes for ultra-long cycle life, Lithium metal anodes and dendrite challenges; Cathode Chemistry: Lithium iron phosphate (LFP) cathodes for grid storage, Layered oxide cathodes (NMC, NCA) and structural chemistry, Spinel cathodes (LMO) for high-power systems, Degradation mechanisms in lithium grid batteries, Thermal stability and safety engineering.

Sodium-Based Grid Batteries: Anode Chemistry: Hard carbon anodes for sodium-ion batteries, Alloy and conversion anodes in sodium systems; Cathode Chemistry: Layered oxide cathodes in sodium-ion batteries, Polyanionic cathodes (NASICON, Prussian blue analogues), Sodium-sulfur battery chemistry, Sodium-nickel chloride (ZEBRA) battery systems.

Flow Battery Chemistry for Long-Duration Storage: Principles of redox flow batteries, Vanadium redox flow battery chemistry, Iron-chromium flow batteries, Zinc-bromine flow batteries, Organic redox flow batteries, Membrane and electrolyte chemistry in flow systems.

Metal-Air and Low-Cost Grid Chemistries: Anode Chemistry: Zinc metal anodes in aqueous batteries, Iron metal anodes for grid storage; Cathode Chemistry: Oxygen reduction and oxygen evolution reactions, Catalysts for air electrodes, Zinc-air battery chemistry, Iron-air battery chemistry.

Emerging Large-Scale Battery Technologies: Liquid metal battery chemistry, Aluminum-based batteries for grid storage, Magnesium and calcium battery systems, Organic and polymer battery materials, Hybrid battery energy storage systems, Future directions in ultra-low-cost grid battery chemistry.

Text books

1. Electrochemical Energy Storage for Renewable Sources and Grid Balancing, Patrick T. Moseley, Elsevier.
2. Battery Systems Engineering, Christopher D. Rahn and Chao-Yang Wang, Wiley

Reference books

3. Grid Energy Storage, U.S. Department of Energy (DOE).
4. Energy Storage Technologies and Applications, Ahmed Faheem Zobaa.

Title	Batteries for Portable Electronics	Number	MTL7XX0
Department	Materials Engineering	L-T-P [C]	3-0-0 [3]
Offered for	M.Tech.(BMT) / M.Sc. / B.Tech. (Final Year)	Type	E
Prerequisite			

Objective:

1. Understand advanced electrochemical mechanisms, fundamental electrochemistry, and material chemistries used in batteries for portable applications.
2. Study cell chemistries, electrode materials, microbattery architectures, micro-fabrication techniques, battery management systems, and emerging technologies for portable electronics and miniaturized systems.

Learning Outcomes:

1. Describe the role, requirements, and historical evolution of batteries used in portable and miniaturized applications.
2. Compare and evaluate different battery chemistries used in portable electronics and microbattery systems.

Contents:

Introduction to Portable Energy Storage: Evolution of portable electronic devices and energy storage needs, Role of batteries in portable electronics, Power and energy requirements of portable devices, Performance metrics: energy density, power density, cycle life, Comparison of primary and secondary batteries for portable devices, Safety and reliability considerations

Battery Chemistries Used in Portable Electronics: Overview of common battery chemistries, Alkaline batteries for portable electronics, Zinc-carbon batteries, Nickel–cadmium (Ni-Cd) batteries, Nickel–metal hydride (Ni-MH) batteries, Lithium-ion batteries, Lithium-polymer batteries, Emerging chemistries for portable applications

Electrode Materials and Cell Chemistry: Fundamentals of electrochemical reactions in portable batteries, Cathode materials for Li-ion batteries ($LiCoO_2$, NMC, LFP), Anode materials (graphite, silicon, lithium titanate), Electrolytes used in portable batteries, Separators and current collectors, Solid-state and gel polymer electrolytes, Interfacial reactions and degradation mechanisms

Battery Design and Cell Formats: Cell configurations for portable devices, Cylindrical cells, Prismatic cell, Pouch cells, Thin-film batteries, Flexible and wearable batteries, Miniaturized battery design

Battery Management and Performance: Charging methods for portable batteries, Battery Management Systems (BMS) for portable electronics, Fast charging technologies, Thermal

management in small batteries, State-of-charge (SOC) and state-of-health (SOH) estimation, Battery aging and cycle life

Applications and Future Technologies: Batteries in smartphones and laptops, Batteries for wearable electronics and smart devices, Batteries for medical portable devices, Integration with energy harvesting systems, Future trends: solid-state batteries and ultra-thin batteries

Text books

1. Batteries in Portable Electronic Devices, Gianfranco Pistoia, Elsevier
2. *Handbook of Batteries* – David Linden, McGraw-Hill

Reference books

1. *Lithium-Ion Batteries: Fundamentals and Applications* – Yuping Wu
2. *Rechargeable Batteries: Materials, Technologies and Applications* – Zhengcheng Zhang

Title	Stationary Batteries & Battery–Grid Integration	Number	MTL7XX0
Department	Materials Engineering	L-T-P [C]	3-0-0 [3]
Offered for	M.TECH.(BMT) / M.Sc. / B.Tech. (Final Year)	Type	E
Prerequisite	Basic Electrochemistry / Materials Science		

Objective:

1. Provide an in-depth understanding of electrochemical energy storage systems, including advanced battery chemistries, cell engineering, and system engineering for stationary and grid-scale applications.
2. Study grid integration strategies, power electronics interfaces, large-scale battery management systems, safety regulations, and emerging technologies for sustainable and reliable grid energy storage.

Learning Outcomes:

1. Compare the characteristics and performance of advanced battery chemistries including lithium-ion, sodium-ion, redox flow, lead-acid, and metal-air systems.
2. Critically evaluate the suitability of these battery chemistries for grid-scale energy storage applications.

Contents:

Introduction to Stationary Energy Storage: Role of energy storage in modern power systems, Overview of stationary battery technologies, Differences between mobile and stationary batteries, Energy storage requirements for power grids, Grid stability, reliability, and resilience, Key performance metrics for stationary storage (energy capacity, power rating, lifetime)

Battery Technologies for Stationary Applications: Lead–acid batteries for grid storage, Lithium-ion batteries for stationary storage, Sodium–sulfur batteries, Sodium–nickel chloride (Zebra) batteries, Flow batteries (Vanadium redox, zinc–bromine), Metal–air batteries for large-scale storage, Molten metal batteries, Emerging stationary battery chemistries

Materials and Electrochemistry: Electrochemical fundamentals of stationary batteries, Cathode materials for large-scale batteries, Anode materials and current collectors, Electrolytes for stationary systems, Membranes and separators in flow batteries, Degradation mechanisms in stationary batteries, Thermal stability and safety considerations

Battery System Design and Battery Energy Storage Systems (BESS): Architecture of Battery Energy Storage Systems, Battery pack design for stationary storage, Battery management systems (BMS) for grid batteries, Thermal management for large battery systems, Power electronics and inverters, Grid-scale battery installation and infrastructure, Safety systems and protection strategies

Battery–Grid Integration: Grid services provided by battery systems, Frequency regulation and voltage control, Load leveling and peak shaving, Renewable energy integration (solar and wind), Microgrids and distributed energy storage, Smart grids and digital control of battery systems

Economics, Policy, and Future Trends: Economic analysis of grid-scale battery systems, Lifecycle analysis and sustainability, Grid regulations and standards for energy storage, Recycling and second-life batteries for stationary use, Future trends in long-duration energy storage

Text books

1. Electrochemical Energy Storage for Renewable Sources and Grid Balancing, Patrick T. Moseley, Elsevier.
2. Battery Systems Engineering, Christopher D. Rahn and Chao-Yang Wang, Wiley

Reference books

3. Grid Energy Storage, U.S. Department of Energy (DOE).
4. Energy Storage Technologies and Applications, Ahmed Faheem Zobaa.

Title	Traction Batteries: Battery Materials & Pack Design	Number	MTL7XX0
Department	Materials Engineering	L-T-P [C]	3-0-0 [3]
Offered for	M.Tech. (BMT) / M.Sc. / B.Tech. (Final Year)	Type	E
Prerequisite	Basic Electrochemistry / Materials Science		

Objective:

1. Equip students with a thorough understanding of battery materials and cell chemistries used in traction applications.
2. Study battery pack design principles for electric vehicles and heavy mobility systems.

Learning Outcomes:

1. Evaluate advanced electrode and electrolyte materials, compare cell formats and pack architectures, and apply thermal, mechanical, and safety engineering principles for traction battery design.
2. Assess battery management strategies, degradation mechanisms, fast-charging protocols, and emerging chemistries for sustainable and next-generation electric vehicle applications.

Contents:

Introduction to Traction Batteries: Evolution of electric mobility and traction batteries, Role of batteries in electric vehicles (EVs) and hybrid electric vehicles (HEVs), Power and energy requirements for traction systems, Comparison of traction batteries with portable batteries, Performance metrics: energy density, power density, cycle life, safety, Overview of battery technologies used in EVs

Electrochemistry and Battery Materials: Fundamentals of electrochemical energy storage, Cathode materials for traction batteries (LFP, NMC, NCA), High-energy cathodes and advanced cathode materials, Anode materials (graphite, silicon, lithium titanate), Electrolytes used in traction batteries, Solid electrolytes and next-generation materials, Separators and current collectors, Material degradation and aging mechanisms

Battery Cell Design for EV Applications: Cell architectures for traction batteries, Cylindrical cells for EVs, Prismatic cells for EVs, Pouch cells for EV applications, High-energy vs high-power cell design, Fast-charging cell chemistry, Thermal behavior of traction cells

Battery Pack Design and Engineering: Battery module architecture, Battery pack configuration and layout, Electrical interconnections and busbars, Thermal management systems for EV batteries, Mechanical design and vibration resistance, Battery enclosure and structural integration, Safety design and protection mechanisms

Battery Management Systems (BMS): Functions of BMS in traction batteries, State of charge (SOC) estimation, State of health (SOH) monitoring, Cell balancing techniques, Thermal monitoring and control, fault detection, and safety control

Testing, Safety, and Future Technologies: Performance testing of traction batteries, EV battery safety standards and regulations, Battery recycling and second-life applications, Fast charging and ultra-high-power batteries, Future technologies: solid-state and lithium-metal batteries

Text Books

1. *Electric Vehicle Battery Systems* – Sandeep Dhameja, Newnes (an imprint of Elsevier / Butterworth–Heinemann)
2. Batteries for Electric Vehicles, Iqbal Husain, IET

Reference Books

- *Handbook of Lithium-Ion Battery Pack Design* – John T. Warner
- *Lithium-Ion Batteries for Electric Vehicles* – Bruno Scrosati

Title	Battery Management Systems	Number	MTL7XX0
Department	Materials Engineering	L-T-P [C]	3-0-0 [3]
Offered for	M.Tech.(BMT) / M.Sc. / B.Tech. (Final Year)	Type	E
Prerequisite	Basic Electrochemistry / Materials Science		

Objective:

1. Provide students with a comprehensive understanding of battery management system architectures and state estimation algorithms for battery packs.
2. Study thermal management strategies for the safe and efficient operation of battery packs in portable, traction, and stationary applications.

Learning Outcomes:

1. Design and evaluate battery management system (BMS) architectures, implement state estimation algorithms (SoC, SoH, SoP), and apply protection and cell balancing strategies for safe battery operation.
2. Analyze battery thermal behaviour, develop thermal management solutions, and integrate BMS with power electronics for optimized battery performance across diverse applications.

Contents:

Introduction to Battery Management Systems (BMS): Role of Battery Management Systems in modern batteries, Functions and architecture of BMS, Battery pack monitoring and protection, Importance of BMS in electric vehicles and stationary storage, Safety challenges in large battery systems, Overview of thermal issues in batteries

Battery Monitoring and Control: Voltage, current, and temperature sensing; Data acquisition systems in BMS, State of Charge (SOC) estimation methods, State of Health (SOH) estimation techniques, State of Power (SOP) estimation, Battery modeling for BMS applications, Algorithms for battery diagnostics

Cell Balancing and Protection: Cell imbalance in battery packs, Passive cell balancing techniques, Active cell balancing methods, Overcharge and over-discharge protection, Short circuit and fault detection, Battery safety protocols and control strategies

Battery Thermal Behavior: Heat generation in batteries, Thermal runaway mechanisms, Thermal modeling of battery cells, Temperature effects on battery performance, Thermal degradation and aging, Safety risks related to overheating

Thermal Management Systems: Air cooling systems, Liquid cooling systems, Phase change materials for thermal control, Refrigerant-based cooling systems, Heat pipes and advanced cooling methods, Thermal design of battery packs

System Integration and Applications: Integration of BMS with thermal management systems, Battery pack architecture and monitoring networks, BMS communication protocols (CAN, LIN, etc.), Control systems for EV battery packs, Testing and validation of BMS and thermal systems, Standards and safety regulations, BMS for stationary battery systems, Future trends: AI-driven battery management and smart thermal systems

Text Books

- *Battery Management Systems* – Gregory L. Plett
- *Battery Systems Engineering* – Christopher D. Rahn & Chao-Yang Wang

Reference Books

- *Lithium-Ion Battery Pack Design* – John T. Warner
- Battery Management Systems: Design by Modelling, H. J. Bergveld and collaborators, Springer

Title	Battery Circular Economy	Number	MTL7XX0
Department	Materials Engineering	L-T-P [C]	3-0-0 [3]
Offered for	M.Tech.(BMT) / M.Sc. / B.Tech. (Final Year)	Type	E
Prerequisite	Basic Electrochemistry / Materials Science		

Objective:

1. Provide students with a thorough understanding of battery degradation mechanisms, maintenance practices, and regeneration techniques for improved battery performance and lifetime.
2. Study second-life reuse strategies and end-of-life recycling technologies for sustainable battery lifecycle management.

Learning Outcomes:

1. Diagnose battery degradation, apply maintenance and regeneration techniques to extend battery life, and evaluate second-life reuse strategies for retired battery packs.
2. Assess hydrometallurgical, pyrometallurgical, and direct recycling processes, and analyze regulatory frameworks and sustainability metrics for end-of-life battery management.

Contents

Introduction to Battery Lifecycle Management: Battery lifecycle: manufacturing to end-of-life, Importance of maintenance and lifecycle management, Battery degradation and aging mechanisms, Performance metrics for battery health, Environmental impact of battery disposal, Circular economy in battery technologies

Battery Maintenance and Diagnostics: Battery inspection and maintenance strategies, Monitoring battery health and performance, State of Charge (SOC) and State of Health (SOH) analysis, Fault detection and diagnostics in battery systems, Thermal management during battery operation, Predictive maintenance using data analytics, Maintenance practices for EV and stationary batteries

Battery Regeneration and Refurbishment: Principles of battery regeneration, Reconditioning of lead–acid batteries, Electrochemical rejuvenation methods, Chemical and additive-based regeneration techniques, Regeneration of lithium-ion batteries, Performance recovery and limitations of regeneration, Economic viability of battery refurbishment

Battery Reuse and Second-Life Applications: Concept of second-life batteries, Evaluation of remaining useful life, Repurposing EV batteries for stationary storage, Testing and sorting for second-life batteries, Design of second-life battery systems, Safety considerations for reused batteries, Case studies of second-life battery applications

Battery Recycling Technologies: Overview of battery recycling processes, Mechanical recycling techniques, Pyrometallurgical recycling methods, Hydrometallurgical recycling processes, Direct

recycling and cathode recovery, Recovery of critical materials (Li, Co, Ni, Mn, Pb), Recycling of different battery chemistries

Policy, Sustainability, and Future Trends: Regulations and policies for battery recycling, environmental impact, and lifecycle assessment, Economics of battery recycling and reuse, Emerging recycling technologies and innovations, Sustainable battery materials and circular battery economy

Text books:

1. *Lithium-Ion Battery Recycling and Reuse* – Y. P. Wu
2. Battery Reference Book, Thomas R. Crompton, Newnes / Elsevier

Reference Books:

3. *Handbook of Battery Materials* – Claus Daniel & Jürgen Besenhard
4. *Lithium-Ion Batteries: Science and Technologies* – Masaki Yoshio

Title	Battery Lifecycle Assessment	Number	MTL7XX0
Department	Materials Engineering	L-T-P [C]	3-0-0 [3]
Offered for	M.Tech.(BMT) / M.Sc. / B.Tech. (Final Year)	Type	E
Prerequisite	Basic Electrochemistry / Materials Science		

Objective:

The Instructor will

1. Provide students with a systematic understanding of lifecycle assessment (LCA) methodology applied to battery systems.
2. Covering environmental impact quantification, carbon footprint analysis, material criticality, and sustainability evaluation across the entire battery value chain.

Learning Outcomes:

1. Students will be able to conduct cradle-to-grave lifecycle assessments for battery systems, quantify environmental impacts including carbon footprint, energy consumption, and resource depletion across mining, manufacturing, use, and end-of-life stages.
2. They will further interpret LCA results to inform sustainable battery design decisions, compare chemistries on environmental grounds, and engage with evolving regulatory and reporting frameworks governing battery sustainability.

Contents

Introduction to Lifecycle Assessment: Concept of lifecycle thinking in energy technologies, Introduction to Lifecycle Assessment (LCA) methodology, Importance of LCA in battery technologies, Environmental impacts of battery production and use, Overview of global sustainability challenges in energy storage, Framework and stages of LCA

Goal and Scope Definition: Defining objectives of an LCA study, Functional unit and system boundaries, Product lifecycle stages in batteries, Data requirements and quality considerations, Attributional vs consequential LCA, Allocation methods in multi-output systems, Case studies of LCA in battery systems.

Life Cycle Inventory (LCI) Analysis: Raw material extraction for battery materials, Mining and processing of lithium, cobalt, nickel, and graphite, Energy consumption in battery manufacturing, inventory analysis of battery production processes, Data collection methods for LCI, Databases and tools for lifecycle inventory, Uncertainty analysis in LCI

Life Cycle Impact Assessment (LCIA): Environmental impact categories (carbon footprint, water footprint, toxicity), Global warming potential and climate impacts, Resource depletion and critical materials, Ecotoxicity and human toxicity assessment, Energy consumption and emissions analysis

Interpretation of impact assessment results

Lifecycle Assessment of Battery Systems: LCA of lithium-ion batteries, LCA of lead–acid batteries, LCA of sodium-ion and emerging batteries, Environmental impacts of EV battery production, Lifecycle analysis of stationary energy storage systems, Role of battery reuse and recycling in lifecycle sustainability

Circular Economy and Sustainable Battery Design: Design for sustainability in batteries, Battery recycling and resource recovery impacts, Second-life batteries and lifecycle extension, Policy frameworks for sustainable battery production, Carbon footprint reduction strategies in battery manufacturing, Lifecycle optimization in battery systems, Future directions in sustainable battery technologies.

Text Book:

1. Hauschild, M. Z., Rosenbaum, R. K., & Olsen, S. I. (Eds.). (2018). *Life Cycle Assessment: Theory and Practice*. Springer.

Reference Book:

1. Everett, B., Peake, S., & Warren, J. (2021). *Energy Systems and Sustainability: Power for a Sustainable Future* (3rd ed.). Oxford University Press.

NAC Feedbacks from Experts & Response

Name	Affiliation	FEEDBACKS	Action taken
Prof. Shobit Omar	IIT Kanpur	Incorporate the design and development of electrode, electrolyte, and separator materials, along with advanced characterization technique s such as CV, GCD, electrochemical impedance spectroscopy and operando/in-situ methods. Electrochemical modelling, multiphysics simulation, and data driven approaches to predict performance, degradation, and thermal behaviour: include in core	Included Under. "Battery fabrication, Operation and Electrochemical Testing Lab" "Computational Electrochemistry and AI-driven battery Design & Diagnostics" is added in the core bucket"
Dr S. Janakiraman	IIT Indore	Dedicated course: "Characterization Techniques for Battery Materials ." This course could cover important methods such as structural (like XRD, Rietveld refinement), morphological (SEM, TEM, etc.), surface characterization (XPS, EDS, AFM), thermal (TGA, DSC), mechanical (tensile, puncture test), and electrochemical characterization techniques (CV, GCD, EIS),	Included under "Battery Material Synthesis and Characterization Lab" & "Battery fabrication, Operation and Electrochemical Testing Lab"
Prof. Amartya Mukhopadhyay	IIT Bombay	I would prefer having 'Electrochemical modelling and simulation' as a core course (which, I believe, will also include the relevant electrochemical concepts, as well as cell life prediction).	"Computational Electrochemistry and AI-driven battery Design & Diagnostics" is added in the core bucket
Prof. Prabeer Barpanda	IISc. Bangalore	Well Designed Program, proceed as it is.	
Dr.Thirupathi	Scientist APT, GFCL EV Products Ltd., Dahej, Gujarat	Overall, the proposed curriculum provides a strong foundation in battery materials and technologies. Incorporation of the above elements could further enhance the electrochemical fundamentals, materials engineering depth, and industry relevance of the program.	All suggestion are included in the final content
Dr. Anirban Maitra	Specialist (Sr.Manager)-Electrochemistry Cell Technology, Matter Energy	1. Industry-Centric Laboratory Modules: Incorporating advanced cell-fabrication practices—such as dry-room handling, slurry rheology, optimization, formation protocols, and safety	1. Available with the institute course as 2. "Computational Electrochemistry and AI-driven battery Design &

	Pvt. Ltd., Ahmedabad	2. Data Analytics & Battery Modelling: A dedicated elective on batter y modelling, BMS fundamentals, SOH/SOC estimation, and data-driven performance prediction would be beneficial. 3. Electrode Manufacturing and Scale-up: Including process-scale topics such as coating/printing technologies, calendaring optimization, pilot-scale production workflows Elective Courses May Include: (a) Cell-to-module and module-to-pack design and developments (b) Welding/joining technologies (ultrasonic, resistance, and laser-based) (c) Automation, formation & aging protocols	Diagnostics" is added in the core bucket 3. Included under "Computational Electrochemistry and AI-driven battery Design & Diagnostics" is added in the core bucket" "Traction batteries: Battery materials & pack design" is added as an elective course
Dr. M Satish	CECRI	I think the proposed course is well designed with weightage to all the component of battery. I am very happy to be a part of this program in different capacitors as mentioned by you.	
Dr. Brij Kishore	CMERI	You have clearly invested considerable thought and effort into designing something that is both rigorous and practical. Please do not hesitate to reach out if you would find it helpful to discuss any of these points in more detail.	